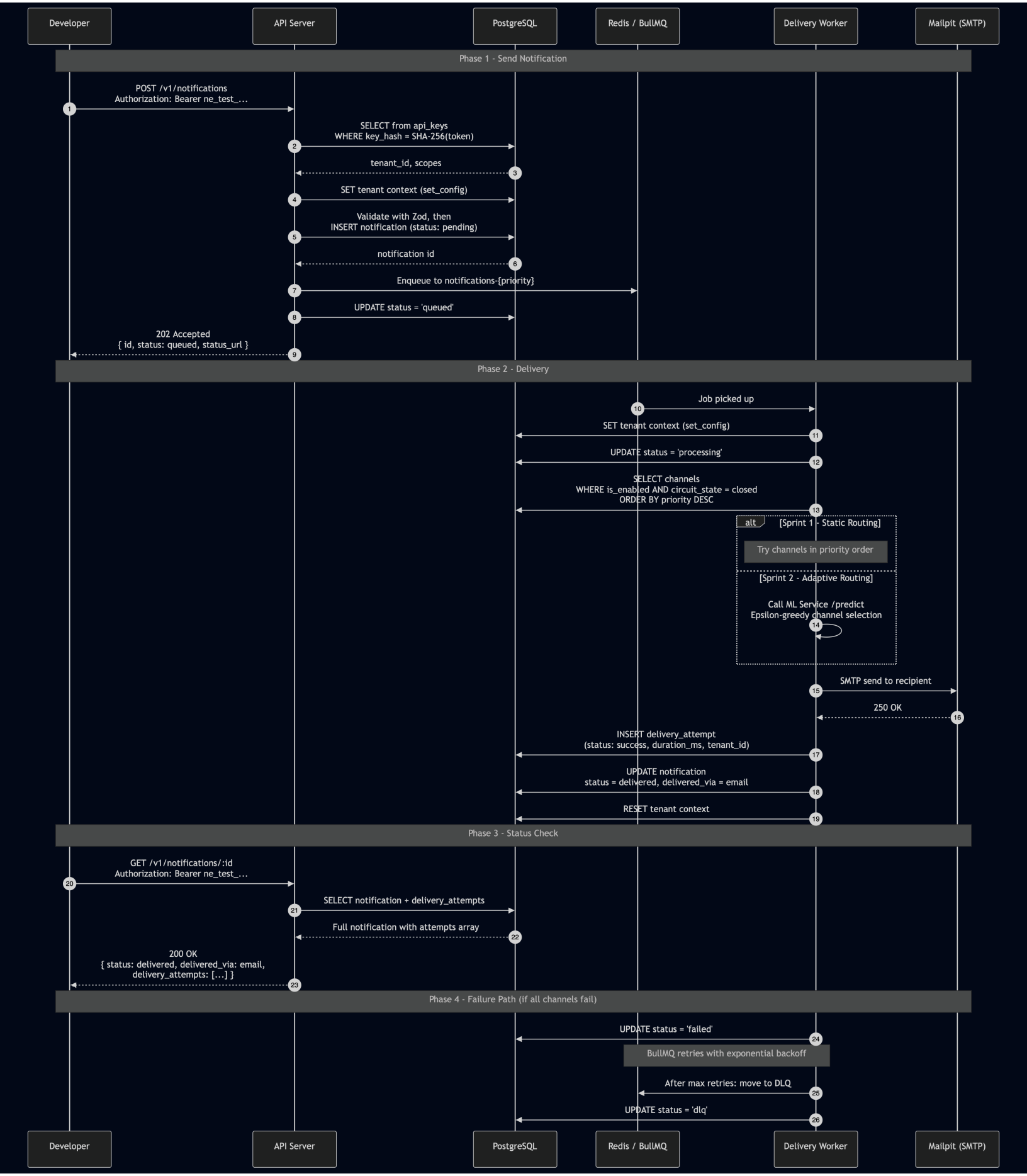
Developer
API Server
PostgreSQL
Redis / BullMQ
Delivery Worker
Mailpit (SMTP)
Phase 1 - Send Notification
1 POST /v1/notifications
Authorization: Bearer ne_test_...
2 SELECT from api_keys
WHERE key_hash = SHA-256(token)
3 tenant_id, scopes
4 SET tenant context (set_config)
5 Validate with Zod, then
INSERT notification (status: pending)
6 notification id
7 Enqueue to notifications-{priority}
8 UPDATE status = 'queued'
9 202 Accepted
{ id, status: queued, status_url }
Phase 2 - Delivery
10 Job picked up
11 SET tenant context (set_config)
12 UPDATE status = 'processing'
13 SELECT channels
WHERE is_enabled AND circuit_state = closed
ORDER BY priority DESC
alt [Sprint 1 - Static Routing]
Try channels in priority order
[Sprint 2 - Adaptive Routing]
Call ML Service /predict
Epsilon-greedy channel selection
14
15 SMTP send to recipient
16 250 OK
17 INSERT delivery_attempt
(status: success, duration_ms, tenant_id)
18 UPDATE notification
status = delivered, delivered_via = email
19 RESET tenant context
Phase 3 - Status Check
20 GET /v1/notifications/:id
Authorization: Bearer ne_test_...
21 SELECT notification + delivery_attempts
22 Full notification with attempts array
23 200 OK
{ status: delivered, delivered_via: email,
delivery_attempts: [...] }
Phase 4 - Failure Path (if all channels fail)
24 UPDATE status = 'failed'
BullMQ retries with exponential backoff
25 After max retries: move to DLQ
26 UPDATE status = 'dlq'
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